2. Connect the Arduino Uno to your computer via the USB port.

Figure 4 shows the circuit design of the Arduino Uno and the Wireless Shield. If everything is connected properly, your Arduino Uno board should look like the one in Figure 5.

The code

When connecting to the internet via Ethernet, we had used the external library *<Ethernet.h>*. Similarly, for connecting the internet wirelessly, we shall be using the external library *<WiFi.h>*.

Note: This library should come preinstalled with your IDE. If for some reason you run into errors, please download this library from the official GitHub repository of the Arduino Project.

```
#include <SPI.h>
#include <WiFi.h>
```

Next, we shall define the constants and variables required for connecting wirelessly. For connecting with the Wi-Fi, we require the *ssid* and *password* of the Wi-Fi that we shall be using. We shall also create a *WiFiClient* variable for connecting to the internet.

```
char ssid[] = "Write WiFi SSID here";
char pass[] = "Write WiFi password
here";

int keyIndex - 0;
int status = WL_IDLE_STATUS;

WiFiClient client;
```

Now we shall define some custom methods for connecting and sending/receiving data over the internet. First up, let us create a method *connectToInternet()* for connecting with the internet.

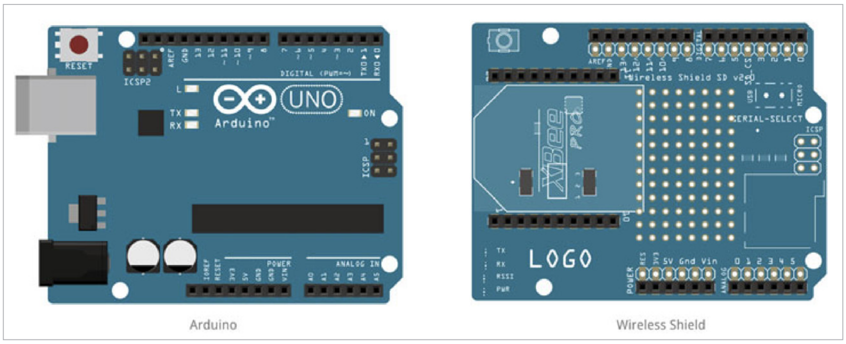

Figure 4: Arduino Uno and Wireless Shield

Figure 5: Wireless Shield attached to Arduino Uno

```
void connectToInternet()
{
    status = WiFi.status();

    if (status == WL_NO_SHIELD)
    {
      Serial.println("[ERROR] WiFi
Shield Not Present");
      while (true);
}

while ( status != WL_CONNECTED)
{
    Serial.print("[INFO] Attempting
Connection - WPA SSID: ");
    Serial.println(ssid);

    status = WiFi.begin(ssid, pass);
}

Serial.print("[INFO] Connection
Successful");
Serial.print("");
printConnectionInformation();
Serial.print
ln("-----------------------");
Serial.println("");
}
```

As you can see in the code above, we have called a custom method *printConnectionInformation()* to display the information of our Wi-Fi connection. So let us go ahead and write that method:

```
void printConnectionInformation()
{
    Serial.print("[INFO] SSID: ");
    Serial.println(WiFi.SSID());

    // Print Router's MAC address
    byte bssid[6];
    WiFi.BSSID(bssid);
    Serial.print("[INFO] BSSID: ");

    Serial.print(bssid[5], HEX);
    Serial.print(":");

    Serial.print(bssid[4], HEX);
    Serial.print(":");

    Serial.print(bssid[3], HEX);
    Serial.print(":");

    Serial.print(bssid[2], HEX);
    Serial.print(":");

    Serial.print(bssid[1], HEX);
    Serial.print(":");

    Serial.println(bssid[0], HEX);

    // Print Signal Strength
    long rssi = WiFi.RSSI();
    Serial.print("[INFO] Signal
Strength (RSSI) : ");
    Serial.println(rssi);

    // Print Encryption type
```